

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
6	(a)		No net force (1) Accept resultant of or sum of. Don't accept equal forces. No net moment (1) Accept all or sum of CM = sum of ACM	2			2		
	(b)		Moments taken about hinge. Clockwise moment shown to be 240×2 or 480 seen (1) Anticlockwise moment: $T \sin 35^\circ \times 1.4$ seen (or $\cos 55^\circ$) (1) $T = 597.8 \text{ [N]}$ (1)		3		3	3	
	(c)	(i)	1.5 mm cable (1) Safe working load > than 600 N [at 3:1 ratio] or only just above 600 N so to be safe will use 2 mm cable (1)			2	2	1	
		(ii)	Accept two $\times$ (1) from the following options: • Shopkeeper does not have necessary expertise to decide on appropriate cable to use • Should seek advice of engineer as suggested on web page or expert advice • Shopkeeper should verify/ validate/ confirm/ authenticate wire rope data • Taken rod to be weightless • [If unsure] should have opted for higher diameter or applied higher safer working ratio • Weather/high winds could affect tension • Chosen wire does support the sign so made an informed choice			2	2		
			Question 6 total	2	3	4	9	4	0